SMART INDIA HACKATHON 2026

Problem Statement ID - 26231

Problem Statement Title -Digital Companion for Field Drug Testing

Theme - MedTech / BioTech / HealthTech

PS Category - Software

Team ID - K26234

Team Name - HexTech

HexTech

Proposed Solution

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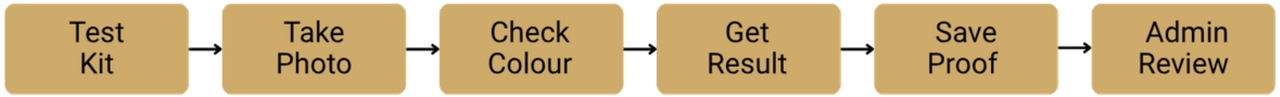
Problem

- Officers check the colour change of drug-test kits by eye. Different officers or lighting can lead to different results.
- There is no fixed reference or number to compare the colour against. Every officer decides on their own.
- There is no proper digital record of when, where, and by whom the test was done, making the result difficult to verify later.

Solution

- Mobile app captures the colour change and checks the test result automatically.
- Records the result, time, location, and Inspector ID for future verification.
- The admin dashboard manages Inspectors, test records, evidence, and reports.

Our Approach



Kit profile (stored as data): demo Marquis kit = purple (98,45,120) · orange (178,94,40) · no change (236,232,214), saved in CIELAB, with limits ΔE2000 ≤ 14 and a 4-point lead. New kits are added from the Admin website.

Novelty

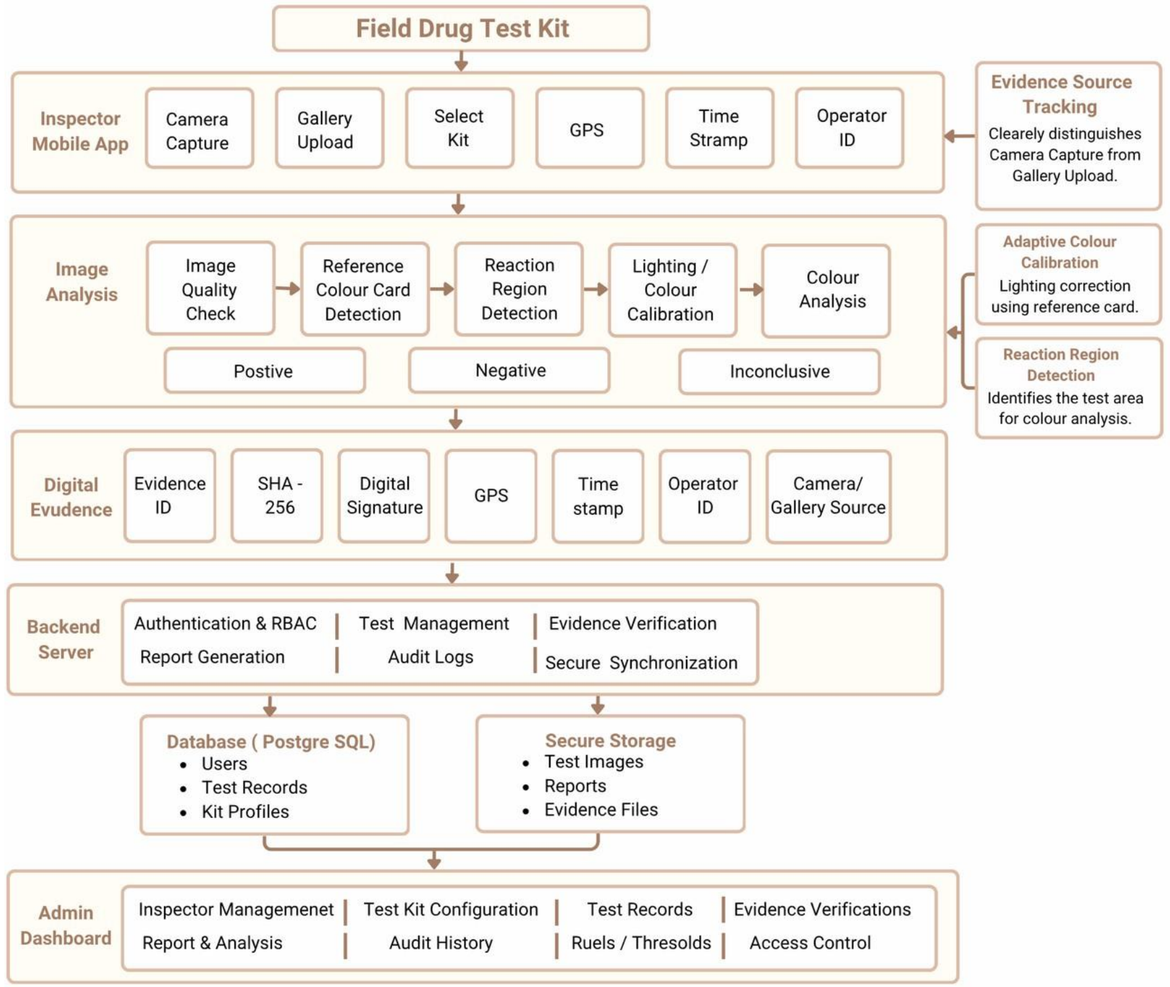
- App reads the colour, not the eye – A low-cost colour card helps handle lighting, and the app gives a colour-based result.
- Honestly says “Not Sure” – Blurry, shiny, or unclear images give an INCONCLUSIVE result with a reason instead of guessing.
- Secure proof of the test – Photo hash, GPS, internet time, and Inspector ID are digitally signed and can be verified using a QR code.
- Works with different test kits – New kit colour profiles can be added from the Admin website without updating the app.

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TECHNICAL APPROACH

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System Architecture



How It Decides & Legal Evidence

1 Find 4 ArUco corners → straighten the card

2 24 known colours → 3×4 colour fix (least squares)

3 ΔE2000 ≤ 14 + 4-pt lead → POS / NEG, else INCONCLUSIVE

4 SHA-256 photo hash → BSA 2023 s.63 certificate

5 Internet time + GPS + Inspector ID, all signed

6 Ed25519 seal + hash chain → no edit / delete

7 QR verify (5 checks) BNSS s.105 · IT Act s.3

Technology Stack

Technology	Purpose
Flutter (Dart)	Mobile App
FastAPI (Python)	Backend
OpenCV + NumPy	Image & Colour Processing
CIELAB + ΔE2000	Colour Matching
PostgreSQL + SQLAlchemy	Database
React + Vite	Admin Dashboard
JWT + bcrypt	Authentication & Roles
SHA-256	Photo Fingerprinting
Ed25519	Digital Signature
NTP	Trusted Time
Leaflet + OpenStreetMap	Map
ReportLab	PDF Reports

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FEASIBILITY AND VIABILITY

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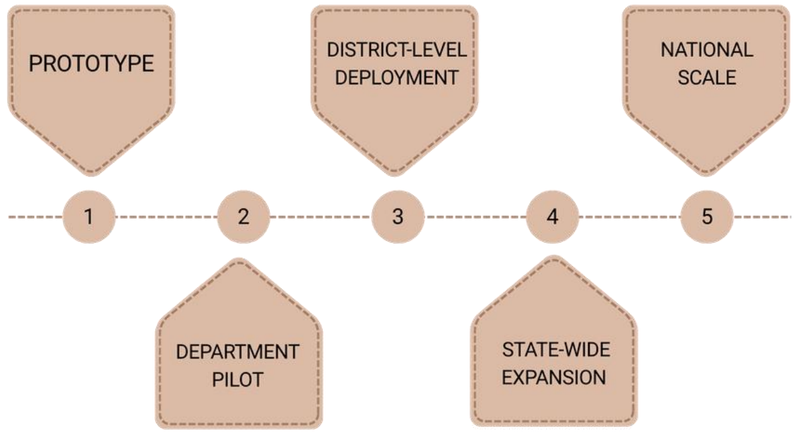
Feasibility & Sustainability

- **Technical –** built on our tested code + OpenCV; runs on any Android phone.
- **Easy to use –** one photo, result in seconds; new kits added from the website.
- **Cost –** ₹10 reusable card + ~₹3,000/month server → under ₹1 per test (est.)
- **Upkeep –** owned by NCB / State ANTF IT cell; FSL updates kit colours per batch; open-source, no licence fee.
- **Data retention –** records kept till the case + appeal end; role-based access; photo shows only card + kit (DPDP 2023).

Potential Challenges & Strategies

Real-world problem	What our app does
Bad light (street lamp / torch)	Colour card fixes the colours
Shiny or blurry photo, card missing	Says NOT SURE and asks to retake
Colour is in between two results	Says NOT SURE; senior officer checks
Fake photo or edited record	In-app camera only + photo fingerprint
Different kits and phones	Kit settings + colours fixed every photo

Phased Implementation & Scalability



Our Goals – Measured Results (tested 28 Sep 2026)

- **Dataset –** 37 test photos in 5 lights + 6 real phone photos.
- **Accuracy –** 25/25 right: 5/5 each in daylight, tungsten, cool, dim and street lamp. Real photos: 5/6 right, 1 NOT SURE, 0 wrong.
- **Decision limits –** ΔE ≤ 14 + 4-pt lead (demo kit); card fit ≤ 9; glare ≤ 3%.
- **Speed –** result + sealed record in 2.5 s typical, 5 s max (goal: 10 s).
- **Tamper test –** 1 photo byte, result or seal changed → verify FAILS (4/4).

Kit colours are simulated swatches (test photos + card on a screen); real kits need an FSL calibration trial.

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IMPACT AND BENEFITS

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Inspector Benefits

- Faster, more consistent testing
- Less manual colour checking
- INCONCLUSIVE when unclear
- Easy digital test records
- Quick access to past test records

Organisation Benefits

- Proof of every test: time, place, officer & photo
- Secure and verifiable records
- Less paperwork with PDF & QR
- Easy search, reports and monitoring
- Better field-test documentation

QR-Based Record Verification

FieldProof - Field Test Record

Presumptive field result — not a substitute for laboratory confirmatory testing.

POSITIVE
Opium-type drug (purple) | 76% sure



Scan to check this record
http://127.0.0.1:8000/v/11

Test number	11
Test ID	4878cf1d-8f00-4f66-9f73-30c50ed39e62
Saved at (UTC)	2026-09-28T06:58:23.601Z (time from: internet time (HTTPS))
Phone time	2026-09-28T12:28:21+05:30 (difference: -2.6 s)
GPS location	10.935299, 76.743282 (±14.791999816894531 m)
Officer	Officer One (Demo) (user id 1)
Device ID	android-1e42a9c170701656ac8a22b0ca44a6e0
Kit	Marquis test (simulated swatches) [DEMO-MARQUIS-SIM], profile version 1
Lab sample ID	-
Review status	NOT_NEEDED

FieldProof · Record check

✓ Record is genuine
The photo and details were not changed after they were saved.

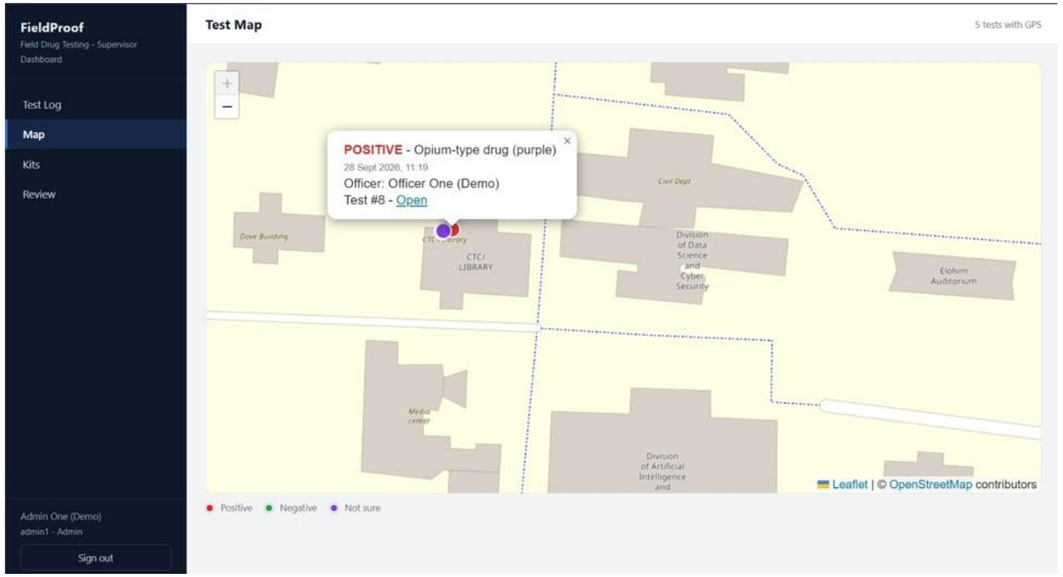
What we checked

- ✓ The photo is the same as when it was taken
- ✓ The saved details match their fingerprint (hash)
- ✓ The FieldProof server's digital signature is valid
- ✓ The record is correctly linked to the record before it
- ✓ The result, officer and kit shown are the ones that were signed

Record details

Test number	11
Result	POSITIVE
What the kit showed	Opium-type drug (purple)
How sure	76%
Kit	Marquis test (simulated swatches) (version 1)
Officer	Officer One (Demo)
Saved at (UTC)	2026-09-28T06:58:23.601Z (internet time (HTTPS))
GPS	10.93530, 76.74328
Test ID	4878cf1d-8f00-4f66-9f73-30c50ed39e62
Photo fingerprint (SHA-256)	294b35454c7e963c7e0f939b4f0e9cd10a279e69fca62634d61ea030d98a9e57

Map - Geo-Tagged test result



Future Scope (not in this demo)

- Works offline - at highways / borders, sends data later
- Phone security chip - phone also signs every test
- Link with police systems - e-Sakshya and CCTNS
- Match with lab results - find which kits are most accurate
- Hindi + regional languages - with voice help

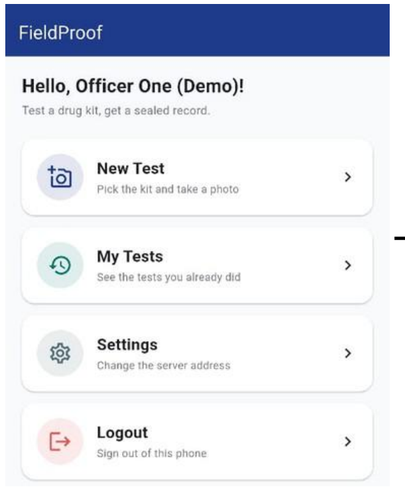
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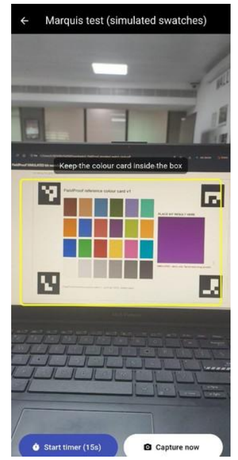
RESEARCH AND REFERENCES

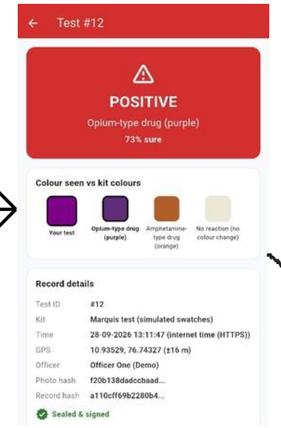
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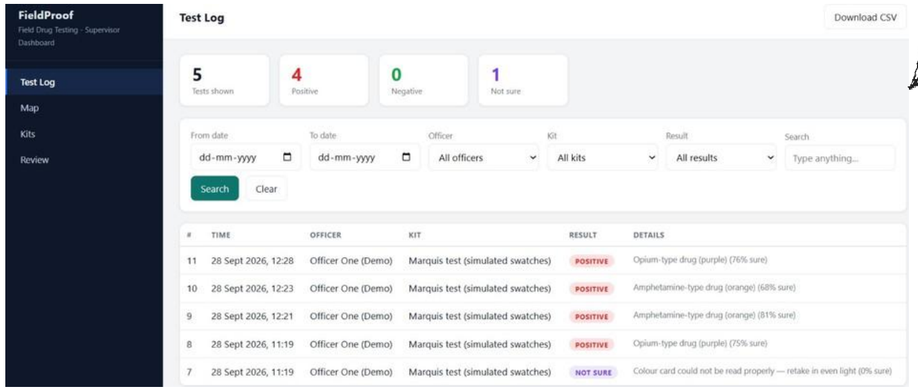
Research	What they found	How we use it
IEEE EMBC 2016 IEEE EMBC 2016 – Illumination and device independence for colorimetric detection of urinary biomarkers with smartphone	Phone camera colours change with light and phone model; reference colours can help handle this.	Our ₹10 colour card fixes colours in every test
IEEE TPS-ISA 2024 IEEE TPS-ISA 2024 – Digital Evidence Chain of Custody : Navigating New Realities of Digital Forensics	Digital evidence must maintain integrity and show how it was handled.	Photo fingerprint + signature + chained log
Forensic Science International 2014 Forensic Science International 2014 – Real-time quantitative colorimetric test for methamphetamine detection using digital and mobile phone technology	A mobile phone camera can measure drug-test colours using RGB values.	Our app reads the test colour as a number
Analytical Chemistry 2016 Analytical Chemistry 2016 – Objective Method for Presumptive Field-Testing of Illicit Drug Possession Using Centrifugal Microdevices and Smartphone Analysis	By-eye colour interpretation is subjective; smartphone image analysis can provide objective analysis.	App gives POSITIVE / NEGATIVE / NOT SURE
Digital Discovery 2023 Digital Discovery 2023 – Teaching old presumptive tests new digital tricks with computer vision	Computer vision can quantify and compare colour changes that may look similar by eye.	App shows photo colour next to kit colours

Prototype Flow – real screenshots of our working build; only kit colours are simulated









Prototype Demo Video: Prototype Video Link